

PyGameMaker — Tutorial 8: Lunar Lander

— Answer Key

For teachers only do not hand out to students. The numbering below matches this tutorial's Worksheet.

Worksheet Answer Key

Part A: 1-C, 2-B, 3-D, 4-A. **Part B:** 1. Create (Set gravity). 2. Keyboard: Up Arrow (held). 3. Collision with obj_pad. 4. Collision with obj_ground (with Restart room). 5. Draw (on the controller). **Part C:**

- 1 The vertical speed stays at about -2 and gravity adds only 0.05 each step, so it takes about 40 steps for the climb to stop. That is why the lander keeps rising before it falls.
- 2 No, the speed does not matter in the tutorial's game. To make a hard landing a crash, test the vertical speed in the pad event: if it is above a limit, crash; otherwise land.
- 3 Gravity keeps pushing the lander into the pad, so the collision event fires again and again. Switch gravity off (strength 0) in the pad event.

Parts D and E: completion and reflection.